Physics 468/668: Statistical Physics **Spring 2019**

Lecture 15: Virial expansion

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

15.1 Cluster expansion

The approach developed in the preceding section leads to exact results only if we apply it to the gaseous phase alone. If we aim to extend it to the phenomena of condensed phase, the critical point, we encounter serious difficulties relating to

- summation over l,
- volume dependence of the cluster integral b_l

Let's focus on the gas phase at the moment.

From

$$\frac{N}{V} = \lim_{V\to\infty}\left(\frac{z}{V}\frac{\partial \ln \mathcal{Q}}{\partial z}\right) = \frac{1}{\lambda^3}\sum_{l=1}^{\infty} l b_l z^l \tag{15.1}$$

$$\sum_{l=1}^{N} \frac{v l b_l z^l}{\lambda^3} = 1 \tag{15.2}$$

where $v = V/N$ denotes the volume per particle in the system.

For the ideal gas, we only care about the first term, thus

$$z = \frac{\lambda^3}{v} \quad \text{(ideal gas)} \tag{15.3}$$

Check Eq. 1.5.7 in the textbook.

For the real gas, as v approaches infinity, the limiting value λ^3/v is obtained for z, shows that if z is expanded as a power series in the volume only inverse power series.

$$z = a_1\frac{\lambda^3}{v} + a_2\left(\frac{\lambda^3}{v}\right)^2 + a_3\left(\frac{\lambda^3}{v}\right)^3 + a_4\left(\frac{\lambda^3}{v}\right)^4 \cdots \tag{15.4}$$

Therefore,

$$\begin{aligned} z^2 &= a_1^2\frac{\lambda^6}{v^2} + 2a_1a_2\frac{\lambda^9}{v^3} + (2a_1a_3 + a_2^2)\frac{\lambda^{12}}{v^4}\cdots \\ z^3 &= a_1^3\frac{\lambda^9}{v^3} + 3a_1^2a_2\frac{\lambda^{12}}{v^4} + \cdots \\ z^4 &= a_1^4\frac{\lambda^{12}}{v^4} + \cdots \end{aligned} \tag{15.5}$$

Now we apply Eq. 15.2,

$$\begin{aligned}
1 &\approx \sum_{l=1}^{4} \frac{v l b_l z^l}{\lambda^3} = \frac{v}{\lambda^3}(b_1 z + 2b_2 z^2 + 3b_3 z^3 + 4b_4 z^4) \\
&\approx \frac{v}{\lambda^3}\left[b_1 a_1 \frac{\lambda^3}{v} + b_1 a_2 \frac{\lambda^6}{v^2} + b_1 a_3 \frac{\lambda^9}{v^3} + b_1 a_4 \frac{\lambda^{12}}{v^4} + 2b_2 a_1^2 \frac{\lambda^6}{v^2} + 4b_2 a_1 a_2 \frac{\lambda^9}{v^3} + (4b_2 a_1 a_3 + 2b_2 a_2^2)\frac{\lambda^{12}}{v^4} \right. \\
&\quad \left. + 3b_3 a_1^3 \frac{\lambda^9}{v^3} + 9b_3 a_1^2 a_2 \frac{\lambda^{12}}{v^4} + 4b_4 a_1^4 \frac{\lambda^{12}}{v^4} \right] \\
&= a_1 b_1 + (a_2 b_1 + 2a_1^2 b_2)\frac{\lambda^3}{v} + (a_3 b_1 + 4a_1 a_2 b_2 + 3a_1^3 b_3)\frac{\lambda^6}{v^2} + (a_4 b_1 + 4b_2 a_1 a_3 + 2b_2 a_2^2 + 9a_1^2 a_2 b_3 + 4a_1^4 b_4)\frac{\lambda^9}{v^3}
\end{aligned} \tag{15.6}$$

To satisfy the condition, we obtain the following relation,

$$\begin{aligned}
a_1 b_1 = 1 &\rightarrow a_1 = b_1 = 1 \\
a_2 b_1 + 2a_1^2 b_2 = 0 &\rightarrow a_2 = -2b_2 \\
a_3 b_1 + 4a_1 a_2 b_2 + 3a_1^3 b_3 &\rightarrow a_3 = 8b_2^2 - 3b_3 \\
a_4 b_1 + 2b_2 a_1 a_3 + b_2 a_2^2 + a_1^2 a_2 b_3 + a_1^4 b_4 &\rightarrow a_4 = -40b_2^3 + 30 b_2 b_3 - 4b_4
\end{aligned} \tag{15.7}$$

We now proceed to expand Pv/kT as a series of z and then v,

$$\begin{aligned}
\frac{Pv}{kT} &= \frac{v}{\lambda^3}\sum_{l=1}^{\infty} b_l z^l \\
&\approx a_1 b_1 + (b_1 a_2 + b_2 a_1^2)\frac{\lambda^3}{v} + (b_1 a_3 + 2a_1 a_2 b_2 + a_1^3 b_3)\frac{\lambda^6}{v^3} + \cdots \\
&= b_1 - b_2 \frac{\lambda^3}{v} + (4b_2^2 - 2b_3)\frac{\lambda^6}{v^3} + \cdots \\
&= \sum_{l=1}^{\infty} a'_l(T)\left(\frac{\lambda^3}{v}\right)^{l-1}
\end{aligned} \tag{15.8}$$

This is called *virial expansion* and a_l, b_l are called *virial coefficients*. To determine the relationship between a'_l and b_l, we obtain

$$\begin{aligned}
a'_1 &= b_1 = 1 \\
a'_2 &= -b_2 = -\frac{2\pi}{\lambda^3}\int_0^{\infty}\left(e^{-u(r)/kT} - 1\right) r^2 dr \\
a'_3 &= 4b_2^2 - 2b_3 = -\frac{1}{3\lambda^6}\int\int f_{12} f_{13} f_{23} d^3 r_{12} d^3 r_{13}
\end{aligned} \tag{15.9}$$

Here we note that a'_l is completely determined by the quantities $b_1, b_2, ...b_l$, that is, by the sequence of configuration integrals $Z_1, Z_2, ...Z_l$.

From the above equations, we can find that a'_3 is completely determined by the 3-cluster [1-2-3-1]. This suggests that the higher-ordder virial may be only determined by a special subgroup of l-clusters.

$$a'_l = -\frac{l-1}{l}\beta_{l-1} \quad (l \geq 2), \tag{15.10}$$

where β_{l-1} is the so called irreducible cluster integral, defined as

$$\beta_{l-1} = \frac{1}{(l-1)!\lambda^{3(l-1)}V} \times \text{the sum of all irreducibel } l\text{-clusters} \tag{15.11}$$

Following this, we obtain

$$a'_3 = -\frac{2}{3}\beta_2 \tag{15.12}$$

The quantities β_{l-1} is dimensionless, and closely related to b_{l-1}.

15.2 2nd virial coefficients with the hard sphere potential

If a given system does not depart much from the ideal-gas behavior, its equation of state is given adequately by the first few virial coefficients. Now we can consider the explicit expression of high order terms. For a_2,

$$a'_2 = -b_2 = \frac{2\pi}{\lambda^3}\int_0^{-\infty}(1 - e^{-u(r)/kT})r^2 dr \tag{15.13}$$

Now we consider that $u(r)$ takes a typical semi-empirical formula,

$$u(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right] \tag{15.14}$$

An analysis on $u(r)$ yield that it reaches a minimum of ϵ at a distance $r_0 = 2^{1/6}\sigma$ and rises to an infinitely large for $r < \sigma$, and vanishes to a small (negative) value for$r$,

$$u(r) = +\infty \quad (r < r_0) \tag{15.15}$$

$$u(r) = -u_0(r_0/r^6) \quad (r \geq r_0) \tag{15.16}$$

substituting 15.15 and 15.16 into 15.13, we obtain for the second virial coefficient

$$a'_2 = \frac{2\pi}{\lambda^3}\left[\int_0^{r_0} r^2 dr + \int_{r0}^{\infty}\left[1 - \exp\left(\frac{u_0}{kT}\left(\frac{r_0}{r}\right)^6\right)\right]dr\right] \tag{15.17}$$

The first integral is straightforward; the second one is considerably simplified if we assume that $(u_0/kT) \ll 1$, which makes the integrand very nearly equal to $(u0/kT)(r_0/r)^6$.

$$a'_2 \approx \frac{2\pi r_0^3}{3\lambda}\left(1 - \frac{u_0}{kT}\right) \tag{15.18}$$

Substituting it into the expansion 15.8, we obtain a first-order improvement on the ideal-gas law, namely

$$\begin{aligned} P &\approx \frac{kT}{v}\left\{1 + \frac{2\pi r_0^3}{3v}\left(1 - \frac{u_0}{kT}\right)\right\} \\ &= \frac{kT}{v}\left\{1 + \frac{B_2(T)}{v}\right\} \end{aligned} \tag{15.19}$$

The coefficient B_2, which is also sometimes referred to as the second virial coefficient of the system, is given by

$$B_2 = a'_2\lambda^3 \approx \frac{2\pi r_0^3}{3}\left(1 - \frac{u_0}{kT}\right) \tag{15.20}$$

In our derivation it was explicitly assumed that (i) the potential function $u(r)$ is given by the simplified expressions (3) and (4), and (ii) $(u_0/kT) \ll 1$. We cannot, therefore, expect this formula to be a faithful representation of the second virial coefficient of a real gas. Nevertheless, it does correspond, almost exactly, to the van der Waals approximation to the equation of state of a real gas. This can be seen by rewriting the expression of P in the form

$$\left(P + \frac{2\pi r_0^3 u_0}{3v^2}\right) \approx \frac{kT}{v}\left(1 + \frac{2\pi r_0^3}{3v}\right) \approx \frac{kT}{v}\left(1 - \frac{2\pi r_0^3}{3v}\right)^{-1} \tag{15.21}$$

which lead to

$$\left(P + \frac{a}{v^2}\right)(v - b) \approx kT \tag{15.22}$$

where

$$a = \frac{2\pi r_0^3 u_0}{3} \text{ and } b = \frac{2\pi r_0^3}{3} = 4v_0 \tag{15.23}$$

15.3 B_2 with Lennard Jones potential

A realistic study of the second virial coefficient requires the use of a realistic potential, such as the one given by Lennard-Jones, for evaluating the integral in (1). This has indeed been done and the results obtained are shown in Figure 10.2, where the reduced coefficient $B_2^{'}(= B_2/r_0^3)$ is plotted against the reduced temperature $T'(= kT/\epsilon)$:

$$B_2^{'}(T') = 2\pi \int_0^\infty (1 - e^{-u'(r')/T'} r'^2 dr') \tag{15.24}$$

with

$$u'(r') = \left\{\left(\frac{1}{r'}\right)^{12} - 2\left(\frac{1}{r'}\right)^{6}\right\} \tag{15.25}$$